Cobalt Final Project Installation Instructions

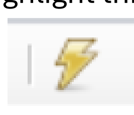
1.Install MySQL Workbench and MySQL Server

To run the application, you must install Database Management Software. We are using MySQL which requires you to install two components:

- MySQL Workbench – GUI to interface with the database
    - Download MySQL Workbench
    - Choose your operating system from the drop-down menu.
    - Following the installation instructions provided.
- MySQL Server – Server for running the database locally on your machine
    - Install MySQL
    - Make sure you select the “Develop Default” option during installation to include the necessary tools.
    - Here is a Video Walkthrough created by John Mattox for a visual guide.

2. Import Original Data into MySQL

The dataset we are going to be using to build out database is from USGS Cobalt Deposits

1. You are going to want to open MySQL Workbench and start the local MySQL server. Make sure you have already installed the FinalProject.
2. Go to File > Open SQL Script.
    - Now navigate to where you are storing the FinalProject Folder and enter the SQL Files folder.
    - From this folder, select “CreateRawDataTables”
3. After opening the file, you will see “CREATE SCHEMA usgs_cobalt_us_project” at the top. Highlight this line of Code and click the lightning bolt icon to run the command.
    - 
    - Now you want to click the refresh arrow next to the Schemas on the left side of your screen.
    - You should now see a schema named usgs_cobalt_us_project.
    - Right click the schema and click “Set as Default Schema”.
4. Now that we are in our new Schema, we are able to create our tables for our raw data. Highlight all of the CREATE TABLE commands and click the lightning bolt.
5. Hit the refresh button again and now you should see all of your new tables underneath the Tables tab.

6. Now we are going to load the CSV Data into our tables. Open the command prompt and use "cd" to navigate to the MySQL binary director.
    - If you are not sure where this is located, open MySQL and then open task manager.
    - If you right-click on MySQL Workbench and select "Open file location" you will see the path to the directory.
    - Example
        - cd "C:\Program Files\MySQL\MySQL Workbench 8.0 CE"
7. Once in the directory type the command:
    - mysql --local_infile=1 -u root -p
    - You will be prompted for your password so enter your unique password you created earlier
8. Now do these commands to get to your database
    - set global local_infile=true;
    - USE usgs_cobalt_us_project
9. Now, go back to MySQL workbench and go to the file explorer to open "LoadCSVDataIntoRowTables.sql"
10. You can see that each LOAD DATA LOCAL INFILE has a path to the .csv file. We need to make sure that this path matches the path to where you are storing the .csv files.
11. All of these .csv files can be found in the Final Project folder under USGS_Cobalt_US_CSV.
12. Go into this folder and find the .csv file that corresponds to each LOAD DATA LOCAL INFILE. Right click on the CSV and click "Copy as Path". Paste this path into the SQL file and make sure to turn any backslashes (\) to forward slashes (/). Also change any " characters surrounding the path to ' charcter
13. Once you have made all of these changes, save the file.
14. To load the data, do Ctrl + A, in the file and press Ctrl + C, to copy all of the contents of the file.
15. Go back to your command prompt that is logged in to the mysql sever and right click to paste all of this code. This should run all the code and load the data into your database.
16. You can check if this worked by going back to your MySQl and clicking refresh like we did before. Right click on a table and press the top option to select rows. There should now be data in the table.

3.Normalize Tables and Insert Data

It is now time to create new tables for our normalized data and to insert the correct data into each normalized table.

1. Click File and now open the SQL Script named "CreateThirdNormalFormTables".
2. Now Do Ctrl+A to select all of the code in the file and press the lightning bolt.
3. This will create all of our new 3rd Normal Form tables.
4. Now, we need to insert our data from the original tables into the correct Normalized Tables.
5. Click File and now open the SQL Script named "InsertOriginalDataIntoNormalizedTables"
6. Do Ctrl-A to select all of the code in the file and press the lightning bolt.
7. Now all of your Normalized tables should be full of data.

4. Load Custom Procedures

We now want to load our custom procedures so that when we run our application, we will have full functionality.

1. Click File and open the SQL Script named "CustomerSpecifiedSQLProcedures".
2. Now Do Ctrl+A to select all of the code in the file and press the lightning bolt.
3. We have now loaded the custom procedures.

5.Install Java and Run the Application

In order to run our application, you must have a Java Extension or Java installed. For the ease of use we are going to use the Java Extension on VS Code

1. Go to [VS Code installation] and click download for whatever operating system you are using.
2. Run the installer and follow the setup prompts.
3. Once the installation has completed, open VS Code and navigate to the extensions button on the left side of the screen.
4. Search for Java Extension Pack by Microsoft and Click install
5. Once this installation has completed, in VS Code go to File > Open Folder and select the folder containing the Java project. In our case this is CobaltProjectApp inside of the FinalProject folder.

6. Once the Project folder has been selected, click on the src folder and you will see CobaltApplication.java inside of it.
7. Click on this Java file and VS Code should detect that it is a Java file and will install anything it is missing.
8. Now click on the file called db.properties.
9. Change the dp.password to the password that you created. Save the file.
10. Now go back to the java file and in the top right corner you will see an arrow that says “Run Java” when you hover over it, click this button and the Java application will launch with a GUI window.
11. You should now have the full functionality of the application

Libraries

This project uses the following open-source libraries:

- Java Swing
    - This is a GUI framework for building the UI of the Cobalt Application.
    - It is part of the standard Java Development Kit